



MINING DEALER BEST PRACTICE

Sprocket Segments Tool for Bolts Installation - D10R, D10N and D10T

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The Installation of the sprocket segments on TTT is a task in which we must be especially careful when tightening the nuts because of the potential safety risks to the technical personnel performing the task. Technicians are exposed to pinch points, arm fatigue, and potential tool slippage.

Currently for the torque procedure of the bolts a lever resting on the final drive is used in order to hold the nuts and ensure correct torque is applied. This exposes the technician to a possible safety issues should the lever slip Fig.1.

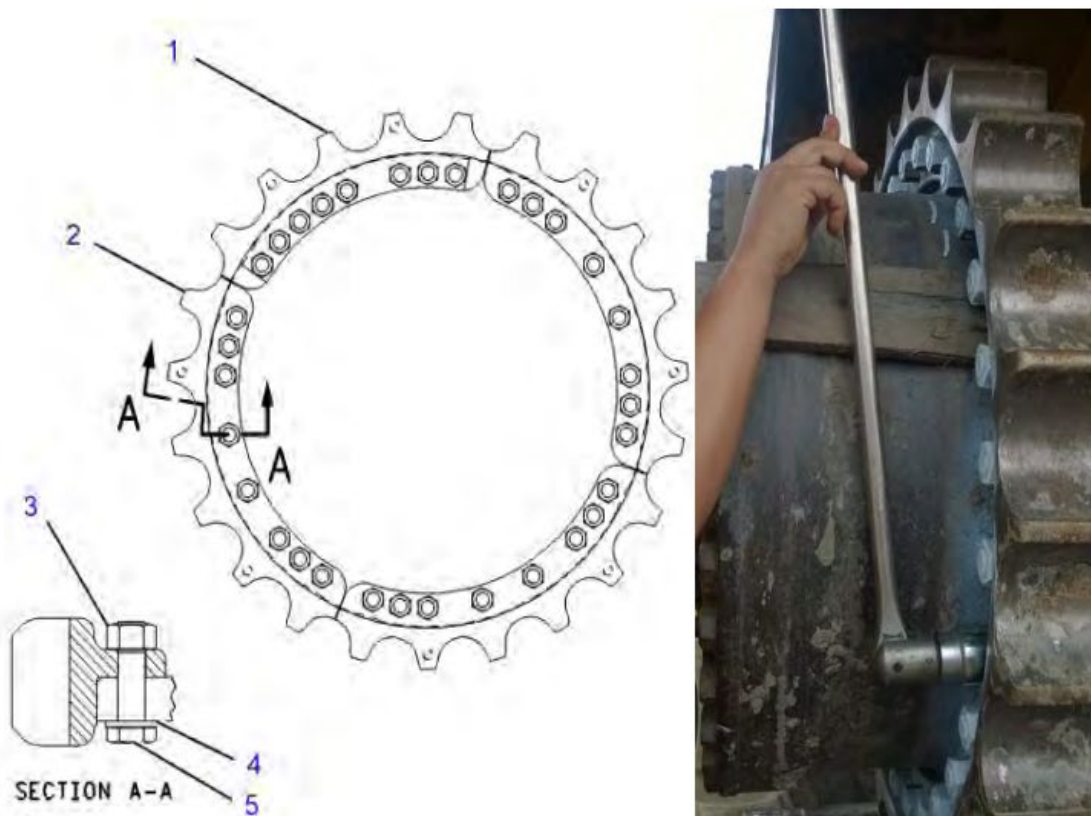


Fig. 1. Former old bolt torque process

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2.0 Best Practice Description

This BP covers the fabrication and use of a simple tool to hold the sprocket segment bolts (D10N, D10R & D10T) replacing the typically used lever, simplifying the process, and reducing the associated risks of sprocket segment changes.

Geometrically, the tool has a "U" shape, in one of its ends, it has a 1 ¼ " and 4 ½" diameter bar in length and in the other end it has a diameter of 2.4" and a length of 1.4" bored concentrically with the silhouette of the head of the bolt to be installed. The two ends are joined by a 1" plate triangle shaped having rounded edges due to the difference in diameters of the two ends. Made entirely of ASTM A36 steel, this tool also has a handle so that the technician can comfortably hold it with his hand when installing or transporting.

In Fig. 4. We can see the way it holds the head of the bolt, and it takes advantage of the segments teeth in order to lock itself and thus it reduces the exposure of the technicians to possible accidents, and it improves the torque of the bolts reducing the installation time by up to 40 %.

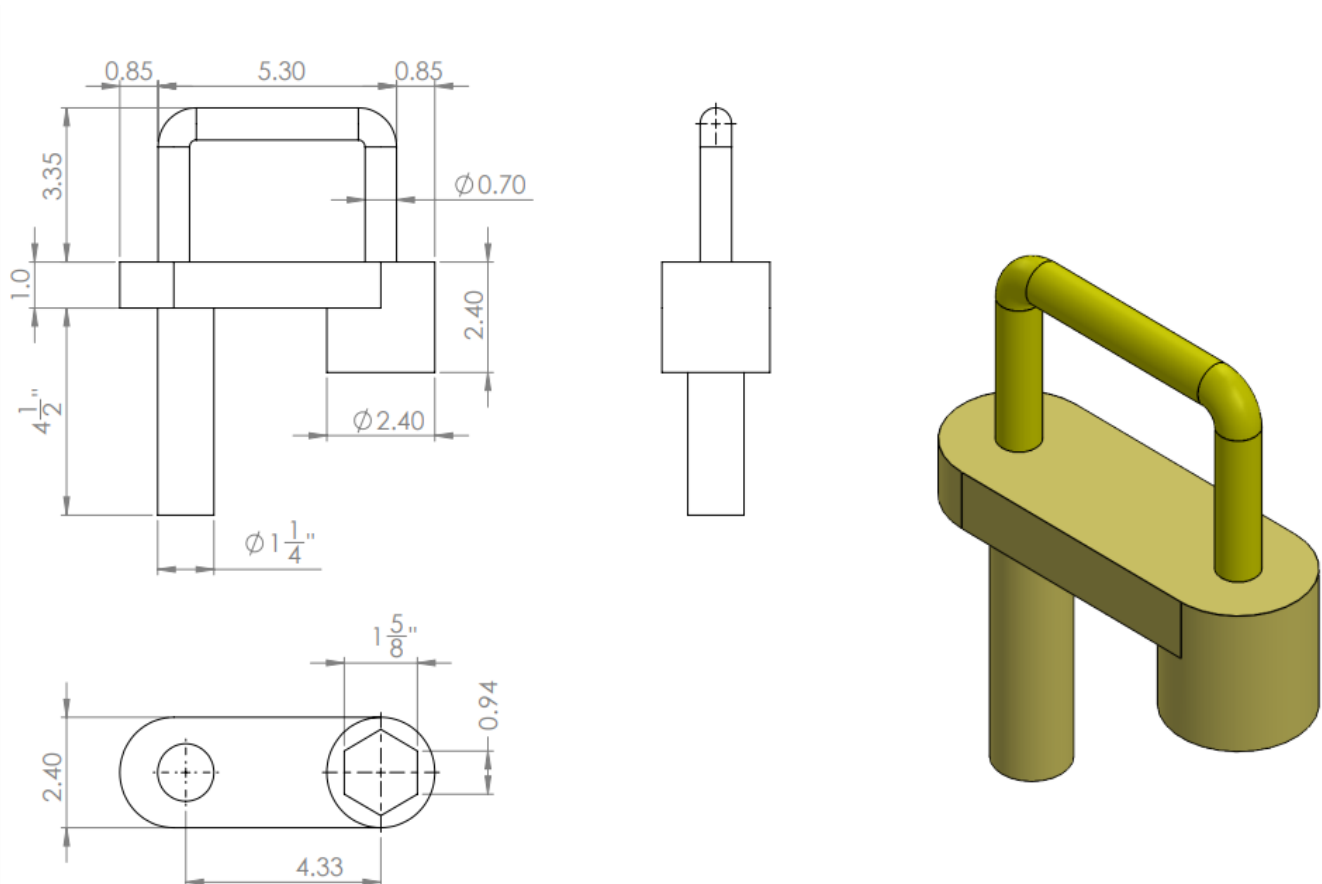


Figure 2. Design drawings, all measurement by inches

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Figure 3. Sprocket segments bolts holding tool

3.0 Implementation Steps

- Manufacturing of the tool.
 - Operation of the tool for applying torque to the screw's segments sprocket (see Figure 4)
1. From the inside of the sprocket, place the largest diameter end of the tool in order to hold the bolt head with the bored silhouette.
 2. Make sure the opposite end of the tool (1 ¼ " bar) is placed in one of the spaces between the sprocket teeth.
 3. install the torque meter along with the adapters required to tighten the nut on the outside of the sprocket, make sure when rotating the tool that it is locked with one of the sprocket teeth thereby preventing the rotation of the bolt.

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Figure 4. Operation of the tool for applying torque the screws segments sprocket

4.0 Benefits

- It improves the safety of the task.
- It reduces the installation time of the sprocket segments up to 40%.
- It improves the torque application.
- There is higher availability of the equipment.

5.0 Resources Required

- Design drawings (see Figure 2)

6.0 Supporting Attachments / References

- SIS: Sprocket Segment - Remove and Install (REN8166-13)

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7.0 Related Best Practices

- Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments

8.0 Acknowledgements

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